

Challenge (Habanero)

- Q1.** A chef is making a quadratic recipe with $x^2 + 5x + 6 = 0$. What is the discriminant?
 (A) 1
 (B) 13
 (C) 25
 (D) 29
 (E) 31
- Q2.** For the quadratic equation $x^2 - 7x - 18 = 0$, how many solutions does it have?
 (A) 1
 (B) 2
 (C) 0
 (D) -1
 (E) -2
- Q3.** A builder is designing a quadratic bridge with the equation $x^2 + 2x - 15 = 0$. What is the quadratic formula solution?
 (A) $x = -5, x = 3$
 (B) $x = -3, x = 5$
 (C) $x = 3, x = -5$
 (D) $x = -2, x = 7$
 (E) $x = 2, x = -7$
- Q4.** For the quadratic equation $2x^2 + 5x + 2 = 0$, what is the value of the discriminant?
 (A) 9
 (B) 17
 (C) 25
 (D) 1
 (E) 9
- Q5.** A quadratic architecture design is modeled by $x^2 - 9x + 20 = 0$. What are the solutions to the equation?
 (A) $x = 4, x = 5$
 (B) $x = 5, x = 4$
 (C) $x = -4, x = -5$
 (D) $x = 4, x = -5$
 (E) $x = -4, x = 5$
- Q6.** The quadratic equation $x^2 + 2x - 6 = 0$ has how many solutions?
 (A) 2
 (B) 1
 (C) 0
 (D) -1
 (E) -2
- Q7.** For the quadratic equation $3x^2 - 2x - 1 = 0$, what is the value of the discriminant?
 (A) 4
 (B) 13
 (C) 16
 (D) 25
 (E) 28
- Q8.** The solutions to $x^2 - 4x - 3 = 0$ can be found using the quadratic formula as $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. What are the solutions?
 (A) $x = -1, x = 3$
 (B) $x = 3, x = -1$
 (C) $x = -3, x = 1$
 (D) $x = 1, x = -3$
 (E) $x = -3, x = -1$

Open Ended Questions (FRQ)

- Q9.** For the quadratic equation $2x^2 + 4x + 1 = 0$, determine the number and type of solutions. Show your work and explain your reasoning.
- Q10.** A chef is making a recipe that involves solving the quadratic equation $x^2 + 3x - 2 = 0$. Use the quadratic formula to find the solutions, and then interpret the results in the context of the recipe. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Emphasize the importance of checking the discriminant before applying the quadratic formula.
- Tip 2: Encourage students to use the quadratic formula to find solutions, even if they can factor the equation.
- Tip 3: Remind students to interpret the solutions in the context of the problem.